

Parth Natekar

IIT MADRAS | UNIVERSITY OF CALIFORNIA, SAN DIEGO

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EDUCATION	INSTITUTION & DEGREE	CGPA	YEAR
	University of California, San Diego <i>MS, Electrical and Computer Engineering</i>	-	2023
	Indian Institute of Technology, Madras <i>Bachelor of Technology in Engineering Design, Master of Technology in Biomedical Design</i>	8.89/10.00	2020
	Fergusson College <i>Higher Secondary School (Major: Electronics)</i>	92.4 %	2015
	Vikhe Patil Memorial School <i>Secondary School (CBSE)</i>	97.0 %	2013
PUBLICATIONS	Yiding Jiang, Parth Natekar, et al. "Methods and Analysis of The First Competition in Predicting Generalization of Deep Learning." <i>Proceedings of the NeurIPS 2020 Competition and Demonstration Track</i>, PMLR 133:170-190 (Link). Parth Natekar, Manik Sharma. "Representation Based Complexity Measured for Predicting Generalization in Deep Learning." <i>Winning Solution of the NeurIPS 2020 competition on Predicting Generalization in Deep Learning</i>, arXiv:2012.02775 (2020) (Preprint). Parth Natekar, Avinash Kori, and Ganapathy Krishnamurthi. "Demystifying Brain Tumour Segmentation Networks: Interpretability and Uncertainty Analysis." <i>Frontiers in Computational Neuroscience</i> 14 (2020): 6 (Link). Avinash Kori, Parth Natekar, Balaji Srinivasan, and Ganapathy Krishnamurthi. "Abstracting Deep Neural Networks into Concept Graphs for Concept Level Interpretability." <i>Best Paper Award, AAAI 2021 Health Intelligence Workshop</i> (Link).		
RESEARCH EXPERIENCE	University of California, San Diego Research Associate 4DCELL Lab (Guide: Dr. Johannes Schoeneberg, UC San Diego) September '21 - Present - Working on Deep Learning for Cellular Imaging and Cell-Organelle Tracking to better understand neuro-degenerative diseases. Researcher ARC Lab & Human-Centered Extended Intelligence Lab (Guide: Dr. Michael Yip & Dr. Nadir Weibel, UC San Diego) June '21 - Present - Working on a joint project between the Advanced Robotics and Controls Lab and the Human-Centered Extended Intelligence Lab on Context Awareness for Augmented Reality Tele-Mentoring. Medical Imaging and Reconstruction Lab, IIT Madras Research Associate Interpretable Deep Learning for Medical Image Analysis (Guide: Dr. Ganapathy Krishnamurthi, Engineering Design, IIT Madras) June '19 - June '20 - Conceptualized a pipeline for Interpretability and Uncertainty Estimation of Deep Learning Models which perform various image processing tasks in the medical domain, such as tumor segmentation. - This involved developing a new method for concept-level interpretability as well as implementing techniques like Network Dissection and Bayesian Dropout on segmentation and classification models. - Implemented the pipeline on benchmark brain tumor segmentation datasets to understand information flow and representation in internal layers and to quantify uncertainty of model predictions. - Developed a python package for direct implementation of the proposed interpretability pipeline on deep neural networks for any relevant biomedical dataset.		

Honeywell Technology Solutions, Bangalore

Research Intern | Graphical Modelling of Neural Networks for Causal Inference

(Mentor: *Satyanarayan Kar*, Honeywell Aerospace, India)

January '19 - June '19

- Devised a probabilistic graphical model based algorithm for Causal Inference on a deep neural network, to provide human-understandable explanations of what the network actually learns.
- Implemented the algorithm on benchmark datasets and successfully delineated variables in the model which were causal for a particular prediction compared to those with spurious correlations.
- Developed a deep learning pipeline for time-series analysis of flight data in order to predict anomalous flight landings. Achieved 86% accuracy in predicting instability 2 minutes prior to landing.
- Implemented an algorithm for object detection and text extraction from aircraft documents.

Convergence Design Lab, Purdue University, USA

Research Intern | Wearables for Human-Computer Interaction

(Guide: *Dr. Karthik Ramani*, Mechanical Engineering, Purdue University) May '18-July '18

- Completed the entire design cycle – ideation, detail design, fabrication and prototyping, of IOT based flexible, wearable electronic devices for interaction with Mixed Reality environments.
- Devised a novel method to design flexible electronic circuits on various fabric substrates. Conducted material and sensor tests to integrate tactile sensing using carbon based piezoresistive sensors.
- Developed multiple ergonomic, working prototypes. Set up networking capabilities via UDP/TCP protocols and tested the wearables in Mixed Reality and IoT environments.
- Created applications in Unity and C# to validate the efficiency of using the prototyped wearables.
- Reduced task time by 44% and error rate by 74% vis-à-vis traditional interaction methods.

CERN (European Organization for Nuclear Research), Geneva

Engineering Intern | Testing Apparatus for CMS cooling system

(Mentor: *Antti Onnela*, EP-DT, CERN, Geneva)

May '17 - July '17

- Selected to be part of a 3-member team from IIT Madras to work on the Large Hadron Collider.
- Designed an adiabatic testing apparatus for the evaporative CO₂ based cooling system of the CMS Detector using concepts from thermal, mechanical and electrical design.
- Set up sensing and feedback system for monitoring of performance of the cooling system
- Created a software interface in LabView to enable easy user interaction.

PROFESSIONAL EXPERIENCE

Anheuser-Busch InBev, Bangalore

Data Scientist | Machine Learning for Compliance Analytics

Aug '20 - May '21

- Worked on building machine learning models and developing hypotheses to flag anomalous and risky transactions for legal and financial compliance.

AWARDS & ACHIEVEMENTS

- **Winner, NeurIPS competition on Predicting Generalization in Deep Learning**, 2020
- **PURE fellow**: Awarded PURE Scholarship for a two-month Research Internship at Purdue University, Indiana, having been selected in the top 14 among 800 students from IIT Madras.
- **CBSE Class X School Topper**: Came first in school with percentage of 97.00%. Awards for highest marks in Math (100/100) and Science (99/100).
- One of 15 students selected from 5000 by Infosys for their '**Spark – Catch Them Young**' Program.

SKILLS

Programming: Python, C, C++, C#, JavaScript, Unity, Machine Learning (Keras, Tensorflow, Pytorch), Arduino, Raspberry Pi, L^AT_EX, SQL
Languages: Competent French, Limited Working German.

RELATED COURSE WORK

Computation & Machine Learning

- Machine Learning for Engineering Applications
- Probabilistic Graphical Modelling
- Statistical Learning
- Computational Systems Biology
- Linear Algebra for Engineers
- Introduction to Computation and Visualization
- Data Structures and Algorithms in Python

Mechanical/Electrical Engineering

- Design of Mechanical Systems (I & II)
- Design of Electronic Systems (I & II)
- Digital Signal Processing
- Engineering Mechanics
- Design of Thermal and Fluid Systems
- Electromagnetic Compatibility for Design
- Bio-Micro-Electrical-Mechanical Systems

Biomedical Design

- Medical Image Analysis
- Design of Diagnostic and Monitoring Systems
- Design of Implantable and Surgical Devices
- Medical Equipment Dissection Lab
- Human Anatomy, Physiology & Biomechanics
- Biomaterials Engineering

Design Methodology & Art

- Functional and Conceptual Design
- Human Factors in Design
- Geometric Modelling and CAD
- Computational Methods in Design
- Design for Form and Aesthetics
- Creative Design

RELEVANT COURSE PROJECTS

Computational Modelling of Pathogen Interactomes

January to May 2018 | Supervisor : [Dr. Karthik Raman](#)

- Delineated drug-resistance pathways in the pathogen Staph Aureus through systems level computational modelling of the protein-protein interaction network of the organism.
- Devised a modified network-based shortest path algorithm by integrating node centrality measures with chemical protein interaction scores between drug molecules and proteins.
- Achieved 68.5% prediction accuracy on already known resistance pathways, validating the viability of our algorithm for exploring drug resistance pathways in pathogens.

Colour Sensing Robotic Manipulator

January to May 2017 | Supervisor : [Dr. N. J. Vasa](#)

- Designed and implemented a 3-dof color detecting robotic manipulator with a magnetic tooltip arm.
- Involved electronic circuit design with the use of a microcontroller to control the robot.
- Coded in C/C++ using the Arduino IDE.

Ultrasound Nerve Image Segmentation

Jan to May 2017 | Supervisor : [Dr. Ganapathy Krishnamurthi](#)

- A deep learning-based project for semantic segmentation of ultrasound nerve cell images.
- Used a fully-convolutional neural-network implemented in Python using Keras and Tensorflow.
- Obtained 71.2% Dice Score on test dataset – on par with then state-of-the-art techniques

TEACHING AND VOLUNTARY EXPERIENCE

Teaching Assistant - Design of Electronic Systems Lab, IIT Madras

Professor : [Dr. Tuhin Subhra Santra](#)

Aug '19 - Dec '19

- Selected as Graduate Teaching Assistant for the Design of Electronic Systems Lab (Fall 2019).
- Managed and guided a group of 60+ students in electronic circuit design principles.

Teaching Assistant - Medical Image Analysis, IIT Madras

Professor : [Dr. Ganapathy Krishnamurthi](#)

Aug '19 - Dec '19

- Conducted tutorial sessions to teach students the programming aspects of Medical Image Analysis.

Volunteering

- Worked with an organization called *Lata Mahajan Foundation* on their project *EducationLinked*, which is aimed at overall development of tribal and underprivileged children.
- Worked with *iVolunteer*, Bangalore to help organize health camps in orphanages.